Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building **data-intensive applications**, **storage layers**, and **web products** across **Python, Flask, React, TypeScript, SQL**, cloud platforms, and proof-of-concept workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant data platform** on Azure SQL and Blob Storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema design, indexing, and storage-layer optimization.
- Built **FastAPI** integration pipelines across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation, reconciliation, and **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform serving **150+ instructors**, handling **10K+ submissions** with secure ingestion, dashboard UX, and API-driven product workflows.
- Designed concurrent backend services on **AWS RDS** with REST APIs and **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, automating build, test, and release flows across **3 operating systems** with **20+ unit/integration suites**.
- Optimized API and ETL workflows using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** resource-management platform with **REST APIs**, WebSockets, and Dockerized microservices, supporting **8+ workflows** and **2M+ synchronized records** across distributed business operations.
- Automated **Jenkins** testing and deployments for Azure SQL-backed services, improving release reliability while reducing resource wastage by **42%** and increasing lead conversion by **15%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a cross-platform booking platform with **React**, TypeScript, **Flask**, and Firebase, supporting **50K+ booking records** with real-time data sync and sub-**200ms** API latency.
- Designed client-facing booking, coupon, and payment flows while mapping **12+ backend integration points**, improving booking confirmation speed by **40%** and reducing manual refunds by **70%**.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Microservices
Frontend: React.js, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Firebase
Cloud & Infrastructure: GCP, AWS (RDS, S3), Microsoft Azure, Docker, Kubernetes, GitHub Actions, Jenkins
Data & AI: Data Warehousing, ETL Pipelines, PyTorch, Transformers, LLM Systems, Feature Engineering
Engineering Practices: Unit & Integration Testing, Performance Optimization, Technical Documentation, Proof-of-Concept Development, Agile

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built a cloud-based document platform over **50K+ documents** with semantic retrieval, collaborative workflows, and **GCP**-backed indexing pipelines that reduced review time by **60%**.

Chat2SQL Live | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, combining Java + TypeScript services, Dockerized deployment, schema exploration, and data-management workflows from a unified UI.

PiXMod Live | *Next.js, CLIP, MongoDB, Stripe*
- Built an AI-powered SaaS product for image processing and semantic search, indexing **100K+ embeddings** with scalable metadata storage and subscription workflows across a React-based frontend.